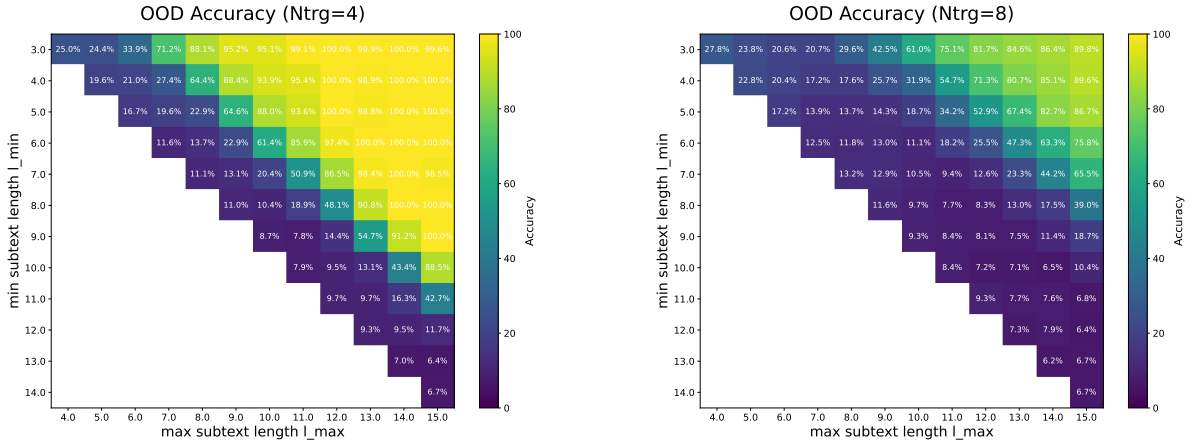


(a) number of triggers $N_{\text{trg}} = 4$

(b) number of triggers $N_{\text{trg}} = 8$

Figure 3: Out-of-distribution accuracy map over varying $\ell_{\min}$ (vertical) and $\ell_{\max}$ (horizontal); moving right indicates greater diversity of ℓ in the pretraining distribution.

- OOD accuracy tends to increase as $\ell_{\max}$ increases (with $\ell_{\min}$ fixed). This suggests that a greater diversity in the training data biases the model towards the induction head.
- Comparing the left and right figures, we see that as the trigger size increases, the region where OOD generalization is achieved shifts rightward, suggesting an increased difficulty of induction head learning with larger N_{trg}, as predicted by Theorem 6.

Error Visualization. Our theory predicts two characteristic error modes:

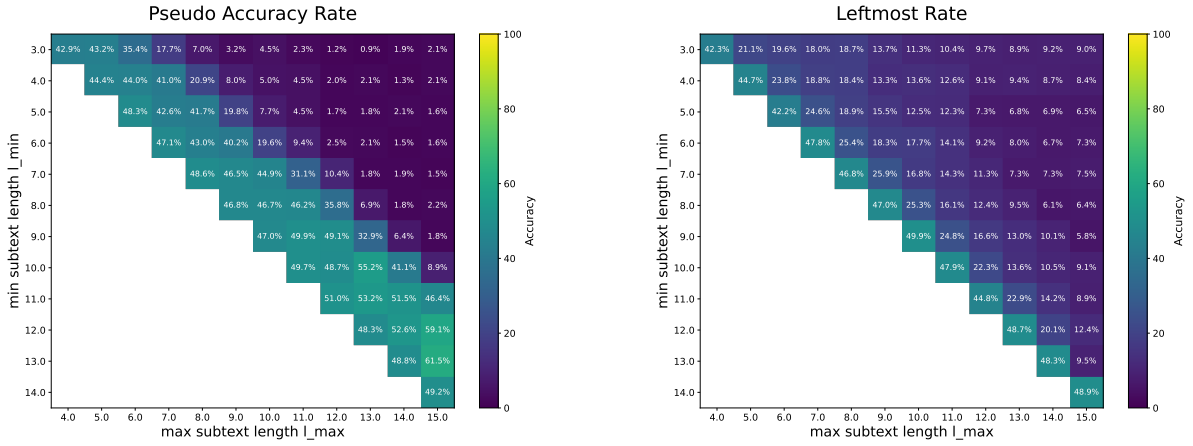
- *Pseudo trigger position.* For non-periodic OOD evaluation data with $\ell_1 + \ell_2 = 2\ell$ and $\ell_1 \neq \ell_2$, let $\tilde{\ell} = (\ell_1 + \ell_2)/2$. The positional shortcut maps the second-trigger position $\ell_1 + \ell_2 + 3$ to the *pseudo* output position $\tilde{\ell} + 2$. Accordingly, we measure the fraction of instances where the model outputs $z_{\tilde{\ell}+2}$ and report this frequency as the *pseudo accuracy rate*.
- *Leftmost position.* Since the leftmost trigger in the pretraining data typically provides the strongest positional signal, the model may output $z_{\ell_{\min}+2}$ *independent of the second trigger position*. This error mode is especially likely when N_{trg} is small. We record its frequency as the *leftmost rate*.

Figure 4 illustrates the existence of these positional shortcuts. We observe that the error rate due to the pseudo-trigger mechanism is higher near the diagonal, and both errors decline as $\ell_{\max}$ increases.

4.2 Experiments for Practical Settings

Next we examine whether a similar transition from positional shortcut to induction head occurs in more standard gradient-based pretraining beyond our theoretical simplification. We consider a three-layer transformer architecture with separated key-query matrices, MLPs, and residual connections, where all parameters are learned *jointly* using the AdamW optimizer [KB14, LH19]. The dataset is generated in the same way as in Section 4.1: we set $N = 32$ and $N_{\text{trg}} = 1$, and varied $\ell_{\min} = 4, 8, \ldots, 20, \ell_{\max} = 4, 8, \ldots, 40$. More experimental details can be found in Appendix E.

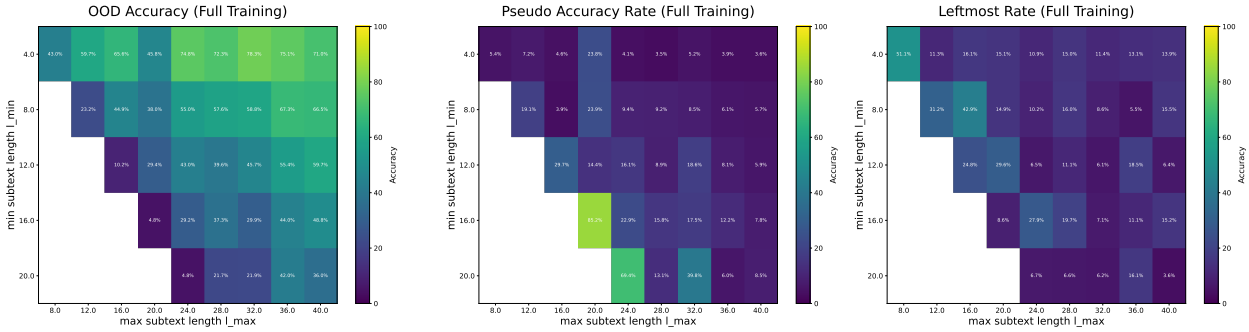
Empirical Observations. Figure 5 shows the OOD accuracy, pseudo accuracy rate, and leftmost rate, following the same setup as in Section 4.1. Note that as the diversity of the pretraining distribution increases, the OOD generalization accuracy improves and the errors due to the positional shortcut decrease — this is consistent with our theoretical prediction in Section 3. We also observe that the transition point is less sharp compared to our theoretical setting.



(a) pseudo accuracy rate ($z_{(\ell_1+\ell_2)/2+2}$)

(b) leftmost rate ($z_{\ell_{\min}+2}$)

Figure 4: Map of two types of errors due to the positional shortcut. Note that both errors can probabilistically coincide with the correct answer, and such cases are not excluded.



(a) OOD accuracy

(b) pseudo accuracy error rate

(c) leftmost error rate

Figure 5: Accuracy map over $\ell_{\min}$ (vertical), $\ell_{\max}$ (horizontal) for a 3-layer transformer trained with AdamW.

5 Conclusion

In this work, using a simplified trigger–output task, we developed a theoretical analysis showing that gradient-based training implicitly selects between two distinct mechanisms with different out-of-distribution generalization properties — an induction head or a positional shortcut. We introduced the *max-sum ratio* as a key quantity governing this selection. Our results demonstrate that the statistical structure of pretraining data critically shapes the algorithms internalized by transformers, offering quantitative insights into steering learning via data design.

We conclude with several directions for future work. First, beyond absolute positional embeddings, it is important to characterize which positional shortcuts can arise under relative position embeddings and related variants. Second, while our analysis centers on a single-layer architecture, a two-layer model naturally delegates retrieval to the first layer (recovering the token corresponding to $\boldsymbol{e}_{z_{t-1}}$); analyzing the coupled dynamics that emerge from this decomposition is an intriguing next step. Finally, developing methods to analyze and quantify richer classes of algorithmic biases – beyond the induction–shortcut dichotomy – would deepen our understanding of how pretraining distributions induce specific computational circuits.

Acknowledgements

RK was partially supported by JST CREST (JPMJCR2115). NN was partially supported by JST ACT-X (JPMJAX24CK) and JST BOOST (JPMJBS2418). TS was partially supported by JSPS KAKENHI (24K02905) and JST CREST (JPMJCR2015). This research is supported by the National Research Foundation, Singapore, Infocomm Media Development Authority under its Trust Tech Funding Initiative, and the Ministry of Digital Development and Information under the AI Visiting Professorship Programme (award number AIVP-2024-004). Any opinions, findings and conclusions or recommendations expressed in this material are those of the author(s) and do not reflect the views of National Research Foundation, Singapore, Infocomm Media Development Authority, and the Ministry of Digital Development and Information.

References

[All25] Zeyuan Allen-Zhu. Physics of Language Models: Part 4.1, Architecture Design and the Magic of Canon Layers. *SSRN Electronic Journal*, May 2025.

[AWKA24] Ekin Akyürek, Bailin Wang, Yoon Kim, and Jacob Andreas. In-context language learning: Architectures and algorithms. *arXiv preprint arXiv:2401.12973*, 2024.

[BCB+23] Alberto Bietti, Vivien Cabannes, Diane Bouchacourt, Herve Jegou, and Leon Bottou. Birth of a transformer: A memory viewpoint. In *Advances in Neural Information Processing Systems*, 2023.

[BEG+22] Boaz Barak, Benjamin Edelman, Surbhi Goel, Sham Kakade, Eran Malach, and Cyril Zhang. Hidden progress in deep learning: Sgd learns parities near the computational limit. *Advances in Neural Information Processing Systems*, 35:21750–21764, 2022.

[BES+22] Jimmy Ba, Murat A Erdogdu, Taiji Suzuki, Zhichao Wang, Denny Wu, and Greg Yang. High-dimensional asymptotics of feature learning: How one gradient step improves the representation. In *Advances in Neural Information Processing Systems 35*, 2022.

[BMR+20] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33, 2020.

[CBKZ24] Hugo Cui, Freya Behrens, Florent Krzakala, and Lenka Zdeborová. A phase transition between positional and semantic learning in a solvable model of dot-product attention. *Advances in Neural Information Processing Systems*, 37:36342–36389, 2024.

[CSWY24] Siyu Chen, Heejune Sheen, Tianhao Wang, and Zhuoran Yang. Unveiling induction heads: Provable training dynamics and feature learning in transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[DLS22] Alexandru Damian, Jason Lee, and Mahdi Soltanolkotabi. Neural networks can learn representations with gradient descent. In *Conference on Learning Theory*, pages 5413–5452. PMLR, 2022.

[ENO+21] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.

[ETE+24] Ezra Edelman, Nikolaos Tsilivis, Benjamin Edelman, Eran Malach, and Surbhi Goel. The evolution of statistical induction heads: In-context learning markov chains. In *Advances in Neural Information Processing Systems*, 2024.